Roll No. | | | | | | | | | | |

Total No. of Pages : 02

Total No. of Questions : 13

B.Pharmacy (Sem.-2)

PHARMACEUTICAL ORGANIC CHEMISTRY-I

Subject Code : BP-202T

M.Code : 74968

Date of Examination : 11-05-2024

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION-A

1. **Briefly write about the following :**

   a) How many isomers are possible for $C_4H_9C_{10}$? Give their IUPAC names.

   b) Define inductive effect. Give suitable example.

   c) What is Markownikoff orientation?

   d) Comment upon the uses of cetosteryl alcohol.

   e) Draw the structure of 2-MethyI-N,N-diethyl butanamine and cylcobutane carboxylic acid.

   f) Define tautomerism. Give example.

   g) Give chemical test to distinguish between 1°, 2° and 3° alcohols.

   h) Give the structure and uses of Amphetamine.

   i) Write any two methods of preparation of alkyl chlorides.

   j) Write down the structure of any two medicinally important alcohols.

Chandigarh

SECTION-B

2. Comment upon following reactions. Give their mechanism and applications if any :

   a) Cannizaro reaction

   b) Benzoin condensation.

3. Give a detailed note on different types of mechanisms involved in the nucleophillic - substitution reactions of alkyl halides. Compare the two in detail and write down the factors affecting these mechanisms.

4. Account for the following :

   a) Electrophillic addition reactions of conjugated dienes.

   b) Halogenation of alkanes.

SECTION-C

5. Explain the effect of substituent on acidity of carboxylic acids.

6. Discuss Kharasch effect taking suitable examples.

7. Write about the effect of base and solvent on El and E2 reactions.

8. What is structural isomerism? Give suitable examples.

9. What are various types of amines? Give examples. How can we distinguish them?

10. Describe various methods of preparation of carbonyl compounds.

11. Give structure and uses of Cetosteryl alcohol and Cinnamaldehyde.

12. Give chemical tests to identify the presence, of alcohols and aldehydes.

13. Comment upon hybridisation in alkanes

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

College of Pharmacy

June-2024